

HIGH-RESOLUTION FEATURE IDENTIFICATION IN HIGH-DIMENSIONAL CLUSTERING

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Interpretable clustering, which involves simultaneous identifying heterogeneous subpopulations and the informative features that define the subpopulations, is a critical yet challenging task across various fields, including omics studies, clinical research, and policy evaluation. Existing methods typically either focus narrowly on global feature heterogeneity or treat feature identification and clustering as separate tasks, failing to account for their interaction. To address these limitations, we propose a novel unsupervised learning approach, PAirwise REciprocal fuSE (PARSE), which concurrently pinpoints cluster-specific informative features and conducts high-dimensional clustering effectively.

1. Introduction. In recent years, data heterogeneity has drawn significant attention across fields such as genomics, precision medicine, marketing, and social science. Cluster analysis, as an unsupervised learning approach, is widely used to uncover heterogeneity patterns by partitioning population into subgroups. Well-studied clustering methods including model-based approaches (Fraley and Raftery, 2002; Handcock et al., 2007; Cai et al., 2019), regularization-based methods (Pan et al., 2013; Ma and Huang, 2017), non-parametric distance-based approaches (Hartigan and Wong, 1979), and spectral-based techniques (Ng et al., 2002; Zhang and Zhou, 2024).

Beyond the traditional focus that is solely on clustering, there is growing interest in a more comprehensive understanding of heterogeneity, addressing both the global partition structure and the identification of partition-specific features. An important example for such a need is clustering in single-cell RNA sequencing (scRNA-seq) analysis, where the primary task is to understand biological heterogeneity by annotating unknown cell types. The standard pipeline includes: (1) clustering cells to identify potential subpopulations and (2) identifying differentially expressed (DE) genes between clusters as potential *cell-type marker genes* (Andrews et al., 2021). To address the challenges of high dimensionality, dimension-reduction procedures, such as principal component analysis in SC3 (Kiselev et al., 2017) and canonical correlation analysis in Seurat (Butler et al., 2018), are commonly applied before clustering. However, Chang (1983) showed that using a reduced number of principal components may fail to preserve clustering information. More critically, the post-clustering differential expression analysis, a key step in benchmark pipelines like Seurat and Scanpy (Wolf et al., 2018), inflates false positives in DE genes detection due to the “double-dipping” issue (Chen and Witten, 2023; Gao et al., 2024; Wang et al., 2024; Yan et al., 2025), as *the genes used for clustering are the same genes tested for differential expression* (Seurat). Several attempts to address this include the truncated normal test (Zhang et al., 2019) and

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the Countsplit method (Neufeld et al., 2024), both of which involve data splitting and assume near-independence of genes; cluster-free DE tests (Vandenbon and Diez, 2020; Chen et al., 2024), which do not identify cell subpopulations; and the ClusterDE method (Song et al., 2023), relying on parametric copulas and lacking theoretical guarantees.

In this paper, we focus on clustering high-dimensional data while simultaneously identifying informative features that define the clusters and recovering the population segmentation pattern for each of these features. We assume the data $\mathbf{y} \in \mathbb{R}^p$, generated from a mixture of elliptical distributions, $f(\mathbf{y}|\boldsymbol{\pi}^*, \mathbf{U}^*, \{\boldsymbol{\Sigma}_k^*\}_{k=1}^K) = \sum_{k=1}^K \pi_k^* f_k(\mathbf{y}|\boldsymbol{\mu}_k^*, \boldsymbol{\Sigma}_k^*)$, where $\boldsymbol{\pi} = (\pi_1^*, \dots, \pi_K^*)^\top$ is the proportion of mixing satisfying $\sum_{k=1}^K \pi_k^* = 1$, and $f_k(\cdot)$ is the density function of the k -th component characterized by its mean vector $\boldsymbol{\mu}_k^* \in \mathbb{R}^p$ and the covariance matrix $\boldsymbol{\Sigma}_k^*$, and matrix $\mathbf{U}_{K \times p}^* = (\boldsymbol{\mu}_1^*, \dots, \boldsymbol{\mu}_K^*)^\top = (\mu_{kj}^*)_{1 \leq k \leq K, 1 \leq j \leq p}$ collects all mean vectors. While $\boldsymbol{\Sigma}_k^*$ may vary across components, homogeneous covariance structures are commonly assumed in high-dimensional model-based clustering (Cai et al., 2019; Mai et al., 2021). This assumption yields linear decision boundaries for separating high-dimensional clusters, which can be modeled via discriminant vectors (Cai and Liu, 2011). In contrast, high-dimensional model-based clustering with heterogeneous covariances typically requires additional structural or regularity assumptions, such as sparsity of the precision matrix (Hao et al., 2017). Accordingly, we develop our method and theoretical guarantees under a homogeneous covariance structure, $\boldsymbol{\Sigma}_k^* = \boldsymbol{\Sigma}^*$. In Section F of the supplementary file, we extend our method to heterogeneous covariance matrices, where additional details are provided. With a latent cluster indicator variable, $z^* \in \{1, \dots, K\}$, the model can be alternatively written as

$$(1.1) \quad \mathbf{y}|\{z^* = k\} \sim f_k(\mathbf{y}|\boldsymbol{\mu}_k^*, \boldsymbol{\Sigma}^*), \quad z^* \sim \text{Multinomial}(\pi_1^*, \dots, \pi_K^*).$$

Motivated by real-world applications, we assume that the cluster structure is driven by a small subset of informative features, while the remaining features present primarily as noises. Including all features can introduce substantial errors into the clustering process.

Conventional feature identification in high-dimensional clustering typically operate at a *low resolution* (Sun et al., 2012; Wang et al., 2018), in the sense that a feature j is selected if it exhibits heterogeneity across clusters, without specifying the pattern of heterogeneity in $\mathbf{U}_{\cdot j} = (\mu_{1j}, \dots, \mu_{Kj})^\top$. These approaches provide unique mean estimates, $\hat{\mu}_{kj}$, for each identified feature across K clusters. In practice, however, individual features, such as cell-type marker genes, often distinguish specific clusters rather than all cluster pairs, as illustrated in Figure 1. That is, population partitions rely on joint contributions from multiple features, while a single informative feature may not distinguish all cluster pairs. Thus, applications like scRNA-seq analysis urge an utmost need for a **high-resolution informative feature map** that (1) identifies the features separating specific cluster pairs and (2) highlights the segmentation patterns for each feature. To this end, we aim to simultaneously perform clustering and identify all non-zero pairwise cluster mean differences, collected in triplets below

$$\xi(\mathbf{U}) = \{(k, k', j) : \mu_{kj} \neq \mu_{k'j}, 1 \leq k < k' \leq K, j = 1, \dots, p\}.$$

Our framework generalizes traditional feature identification, which focus only on identifying the set of informative features: $\mathcal{J} = \{j : (k, k', j) \in \xi(\mathbf{U}) \text{ for some } 1 \leq k < k' \leq K\}$. Fused Lasso or its adaptive variant has been employed to identify significant pairwise differences in cluster means by encouraging sparsity in marginal mean differences between clusters within a Gaussian mixture model (Ma and Huang, 2017; Guo et al., 2010). However, these methods lack theoretical guarantees for either high-resolution feature identification or clustering. To bridge the gap, we propose a novel **PAirwise Reciprocal fuSE (PARSE) penalty** approach, leveraging a reciprocal fusion-type penalty on the pairwise differences among cluster means for each dimension, in a general mixture-of-elliptical-distributions model. In a high-dimensional setting, PARSE simultaneously performs clustering, identifies high-resolution

informative features, and reveals population segmentation patterns for each feature. In addition, we develop an adaptive EM algorithm that uses only intermediate sets of informative features in the E-step. We also establish theoretical guarantees for PARSE and show its optimality in clustering, informative feature identification, and model estimation.

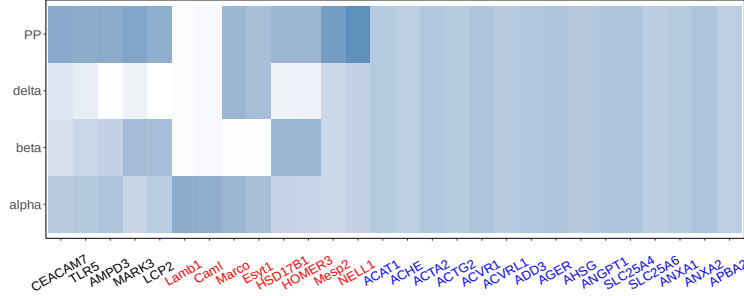


Fig 1: An illustration of four-cell-type scRNA-seq data with informative features that separate all clusters (in black) or part of clusters (in red), and non-informative features (in blue).

1.1. *Our contributions.* Our proposed framework presents several important contributions in methodology, algorithm, and theoretical results. To the best of our knowledge, the proposed PARSE framework is the first to deliver an *interpretable* high-dimensional clustering procedure, enabled by high-resolution feature identification, while ensuring theoretical optimality in clustering, feature identification, and estimation, achieving an oracle property.

Mixture model beyond Gaussian. Most existing model-based clustering methods rely on Gaussian mixture models due to their computational and statistical advantages, particularly in high-dimensional settings (Lindsay, 1995; Fraley and Raftery, 2002; Cai et al., 2019; McLachlan et al., 2019). However, in many applications, the commonly used Gaussian mixture model often fails to adequately capture the underlying data structure (Lin, 2010; Lee and McLachlan, 2013; Forbes and Wraith, 2014). In this paper, we adopt a broader class of elliptical distributions for component densities in a mixture model, allowing us to handle flexible data structures, particularly those with relatively heavy tails. While non-Gaussian distributions in mixture models have been previously explored (Peel and McLachlan, 2000; Azzalini and Valle, 1996; Lee and McLachlan, 2022; Lin, 2010; Maleki et al., 2019), these studies have largely been limited to univariate or low-dimensional cases. High-dimensional applications remain challenging for most existing work, which also lack theoretical guarantees, a gap we address rigorously in this paper.

PARSE penalty for high-resolution feature identification. To identify pairwise informative features responsible for distinguishing specific pairs of clusters, we apply a reciprocal fusion penalty on pairwise cluster means, $\{\mu_{kj} - \mu_{k'j}\}_{1 \leq k < k' \leq K}$ for each dimension j , in corresponding model estimation. Lasso-type penalties, while widely used, often result in over-fitted models under less stringent conditions (Zou, 2006; Zhao and Yu, 2006), producing small non-zero estimates that increase false positives when identifying features specific to certain cluster pairs. In contrast, the reciprocal Lasso penalty, a modified ℓ_0 penalty, heavily penalizes near-zero estimates, promoting sparsity and improving model consistency, as demonstrated in regression settings (Song and Liang, 2015; Freijeiro-González et al., 2022). Equipped with PARSE, our approach simultaneously achieves optimality in clustering, feature identification, and model estimation, fulfilling the oracle property. Though PARSE does not directly provide measures of variable importance, which is beyond the scope of this paper, ad hoc ranking criteria based on the estimated model parameters can still be constructed in practice, as discussed in Section 9.

Adaptive EM algorithm for high-dimensional clustering. The Expectation-Maximization (EM) algorithm is widely used for maximum likelihood estimation in latent variable models. However, because of the non-concavity of the objective function, the EM algorithm typically converges to a local rather than a global maximum (Wu, 1983), whereas statistical guarantees often depend on global optima. Balakrishnan et al. (2017) addressed this issue by explicitly characterizing computational and statistical convergence rates for the EM algorithm under suitable initializations, but their results are limited to low-dimensional settings ($p < n$). In high-dimensional regimes ($p \gg n$), particularly for model-based clustering, even when a regularized approach is computationally feasible (Sun et al., 2012; Witten and Tibshirani, 2012; Chang et al., 2016; Zhao et al., 2018), theoretical guarantees for optimal statistical properties remain elusive due to the influence of high-dimensional nuisance parameters.

A critical challenge in high-dimensional EM algorithm is estimating the inverse of the high-dimensional covariance matrix, particularly necessary for calculating the conditional expectation of latent indicator variables. In high-dimensional settings, the sample covariance matrix is often non-invertible. Existing methods typically address this issue by either assuming a specific covariance structure, such as a diagonal covariance matrix (Wang and Zhu, 2008; Guo et al., 2010), or applying regularization to the covariance or precision matrix (Ruan et al., 2011; Hao et al., 2017; Dong et al., 2024) to facilitate the estimation of conditional expectations. However, despite these computational adjustments, theoretical guarantees regarding the statistical efficiency of regularized high-dimensional EM frameworks remain limited, even with feature identification. Alternatively, Cai and Liu (2011); Cai et al. (2019) adopt a discriminant-analysis-type framework and impose “sparsity” assumptions on the corresponding discriminant factors. However, these methods only work for the Gaussian mixture model and lack the ability to map back to the original features, thus failing to facilitate feature identification and losing interpretability. To address these challenges, we propose a novel EM algorithm that leverages an adaptive set of informative features during iterations, gradually excluding the non-informative features during the E-step. This approach enhances clustering accuracy while bypassing the estimate for the high-dimensional covariance matrix.

Theoretical Contributions. Consistency in estimating cluster means has been studied in the literature (Pan and Shen, 2007; Pan et al., 2013; Wu et al., 2016; Chen and Li, 2009; Xia et al., 2018), yet theoretical guarantees for optimality have largely been absent. Hao et al. (2017) examined the convergence rate of mean estimation in a model-based clustering framework with Lasso. For a two-component mixture model, Jin et al. (2017) derived both the statistical and computational bounds for clustering. Azizyan et al. (2013) established minimax bounds for clustering loss, and Verzelen and Arias-Castro (2017) demonstrated minimax rates for clustering and feature identification in a two-component Gaussian mixture model. However, when the number of clusters exceeds two, the symmetry in cluster means is lost, and theoretical guarantees for feature identification remain overlooked.

We theoretically show that the adaptive posterior probability does not impede the convergence of the proposed adaptive EM algorithm. In fact, it ensures convergence in a high-dimensional regime where posterior probabilities would otherwise be inaccessible without structural assumptions. Upon algorithmic convergence, we establish the statistical guarantees for feature identification, parameter estimation, and clustering. The required minimum signal strength is of the order $n^{(\gamma-1)/4}$ for some $\gamma \in (0, 1)$ and is allowed to converge to zero as n diverges, a much weaker condition than those for traditional mixture models (Cai et al., 2019; Mai et al., 2021). We prove that $\mathbb{P}(\xi(\hat{\mathbf{U}}) \neq \xi(\mathbf{U}^*)) \rightarrow 0$, ensuring consistent and oracle high-resolution feature identification. The identified features possess distinct means across clusters and are conducive to the clustering structure. While low-resolution feature identification consistency has been established for Gaussian mixture models (Sun et al., 2012; Wang et al., 2018), our results are the first to achieve oracle-level feature identification for high-resolution feature identification under a mixture of non-Gaussian distributions.

Furthermore, we establish the convergence rate $O(\sqrt{s_{\max} \log(p)/n})$ for estimating differences among cluster means under ℓ_∞ -norm loss, with model sparsity $s_{\max}$ defined in Section 4. This rate is optimal, matching the lower bound for estimating an s -sparse p -dimensional vector (Johnstone, 1994). Leveraging consistent and oracle feature identification, we formulate the optimal Bayes rule for clustering using the identified features rather than all features, and show that the convergence rate for the mis-clustering error for the estimated clustering assignments relative to the optimal Bayes rule is $O(s_{\max} \log(p)/n)$. This approach relaxes the separation requirements among clusters and addresses the challenge of diminishing signal strength. These theoretical guarantees confirms that PARSE effectively performs clustering while simultaneously identifying high-resolution informative features.

Finally, we examine the fundamental difficulties for feature identification and clustering under the proposed model by carefully constructing a legitimate subspace to cover the parameter space so that the Fano's method leads to the desired lower bounds. Since the Gaussian mixture is as a special case of our non-Gaussian framework, the minimax lower bound is established over a broader parameter space than in previous works (Cai et al., 2019). We show that the minimax risk for feature identification is non-zero, indicating that identifying features in the clustering context is neither simple nor trivial. This underscores the optimality of the proposed method, which can identify the true high-resolution features. Similarly, the minimax convergence rate for clustering assignments relative to the optimal Bayes rule is $O(s_{\max} \log(p)/n)$, matching the upper bound of our proposed algorithm. Thus, the proposed method achieves optimality in both feature identification and clustering simultaneously.

Organization. In Section 2, we introduce a general family of mixture models beyond Gaussian distributions and propose the PARSE penalty for high-resolution feature identification. Section 3 presents our adaptive EM algorithm for simultaneous clustering and feature identification. Sections 4-6 establish the theoretical properties of PARSE, showing its consistency and optimality in identifying globally and pairwise informative features under mild conditions. Section 7 compares PARSE and other popular methods through simulations, while Section 8 highlights its superiority relative to major pipelines in identifying cell subpopulations and cell-type marker genes using human islet cell scRNA-Seq data. Technical proofs and additional numerical results are relegated to the supplement.

2. Methodology. We begin by introducing notation used throughout the paper. Let $\mathbf{1}_n$ denote the n -dimensional vector of ones. For a p -dimensional vector $\mathbf{a} = (a_1, \dots, a_p)^\top$, the ℓ_q -norm is defined as $\|\mathbf{a}\|_q = (\sum_{j=1}^p |a_j|^q)^{1/q}$, $1 \leq q < \infty$. For a $p \times p$ matrix $\mathbf{M}$, let $\lambda_{\min}(\mathbf{M})$ and $\lambda_{\max}(\mathbf{M})$ denote its smallest and largest eigenvalue. Given a vector $\mathbf{a} = (a_j)_{j=1}^p$, a matrix $\mathbf{M} = \{m_{ij}\}_{i,j=1}^p$, and the index set $\mathcal{J} \subset \{1, \dots, p\}$, let $|\mathcal{J}|$ denote the cardinality of $\mathcal{J}$, $\mathbf{a}_{\mathcal{J}} = (a_j)_{j \in \mathcal{J}}$ the corresponding subvector, and $\mathbf{M}_{\mathcal{J}, \mathcal{J}} = \{m_{ij}\}_{i,j \in \mathcal{J}}$ the submatrix of $\mathbf{M}$ indexed by $\mathcal{J}$. For a p -variate density function $f_k(\cdot | \boldsymbol{\mu}_k, \boldsymbol{\Sigma})$, let $f_{k, \mathcal{J}}$ denote the $|\mathcal{J}|$ -variate density with mean $\boldsymbol{\mu}_{k, \mathcal{J}}$ and covariance $\boldsymbol{\Sigma}_{\mathcal{J}, \mathcal{J}}$. For sequences $\{a_n\}$ and $\{b_n\}$, $a_n = o(b_n)$ if $a_n/b_n \rightarrow 0$ as $n \rightarrow \infty$, and $a_n = O(b_n)$ if $\limsup_{n \rightarrow \infty} |a_n|/b_n < \infty$. Similarly, $X_n = o_p(a_n)$ and $X_n = O_p(a_n)$ are defined for random variables X_n . Write $a_n \lesssim b_n$ if and only if $a_n \leq C b_n$ for some C independent of n , and $a_n \asymp b_n$ if $C|b_n| \leq |a_n| \leq D|b_n|$ for constants C, D . Finally, let $\boldsymbol{\pi}^*, \mathbf{U}^*, \boldsymbol{\Sigma}^*$ denote the true population parameters, $\hat{\boldsymbol{\pi}}, \hat{\mathbf{U}}, \hat{\boldsymbol{\Sigma}}$ their estimates, and $\tilde{\boldsymbol{\pi}}, \tilde{\mathbf{U}}, \tilde{\boldsymbol{\Sigma}}$ the corresponding population counterparts of the estimates, with analogous notation for parameter-dependent sets such as $\mathcal{J}$.

2.1. Mixture of elliptical distribution model: Beyond Gaussian. Consider n independent samples of p -variate observations $\mathbf{y}_i = (y_{i1}, \dots, y_{ip})^\top \in \mathbb{R}^p$, for $i = 1, \dots, n$, generated from a K -component mixture model in (1.1). In this paper, we introduce a general *Mixture-of-Elliptical-Distribution* (MoED) model, which encompasses a broad family of distributions.

Specifically, the component density for the k th cluster follows an elliptical distribution

$$(2.1) \quad f_k(\mathbf{y}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) = |\boldsymbol{\Sigma}_k/c_{p,g}|^{-1/2} g\{(\mathbf{y} - \boldsymbol{\mu}_k)^\top (\boldsymbol{\Sigma}_k/c_{p,g})^{-1} (\mathbf{y} - \boldsymbol{\mu}_k)\},$$

which is governed by the mean vector $\boldsymbol{\mu}_k \in \mathbb{R}^p$ and the covariance matrix $\boldsymbol{\Sigma}_k \in \mathbb{R}^{p \times p}$. Let $w(x) = x^{p/2-1}$ on $(0, \infty)$ and define the normalization constant $c_{p,g} = \int_0^\infty x^{p/2} g(x) dx / \{p \int_0^\infty w(x) g(x) dx\}$, where $g \in L_w^1(0, \infty)$, ensuring f_k to be a valid probability density (Fang and Zhang, 1990). To guarantee identifiability of the mixture model, each component distribution f_k is typically assumed to be nearly unimodal (Gupta et al., 2013; Fang et al., 2018), which can be enforced by requiring g to be monotonically non-increasing (Bhattacharyya and Bickel, 2014). We therefore consider the function class $\mathcal{G} = \{g \in L_w^1(0, \infty) \cap W^{1,\infty}(0, \infty) : g' \leq 0 \text{ a.e.}\}$, where $W^{1,\infty}(0, \infty)$ denotes the class of Lipschitz functions with essentially bounded derivative. This refined elliptical family covers a number of important distributions, including the multivariate Gaussian, t -, and Cauchy distributions.

The EM algorithm is widely employed for maximum likelihood estimation in mixture models. If the cluster labels z_i are observed, the negative joint log-likelihood function for the complete data $\mathbf{Y} = (\mathbf{y}_1, \dots, \mathbf{y}_n)$ can be written as

$$(2.2) \quad L_C(\mathbf{Y}, \mathbf{z}, \boldsymbol{\pi}, \mathbf{U}, \{\boldsymbol{\Sigma}_k\}_{k=1}^K) = - \sum_{i=1}^n \sum_{k=1}^K \mathbb{I}(z_i = k) \{\log f_k(\mathbf{y}_i|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) + \log \pi_k\}.$$

The clustering relies on the estimated conditional probabilities, $\mathbb{P}(z_i = k|\mathbf{y}_i, \hat{\boldsymbol{\pi}}, \hat{\mathbf{U}}, \hat{\boldsymbol{\Sigma}})$, which are calculated as the ratio of component likelihoods. Elliptical distributions belong to the location-scale family, and the MoED model serves as a general framework for a wide range of location- and distance-based clustering algorithms (Banfield and Raftery, 1993; Fraley and Raftery, 1998; Balakrishnan et al., 2017).

While the Gaussian mixture model is the most extensively studied framework in model-based clustering, extending it to the general MoED framework is nontrivial, particularly in high-dimensional settings. When $p \gg n$, the conventional EM algorithm may become computationally infeasible, and the maximum likelihood estimator may not even exist. To address this problem, recent methods have integrated a discriminant analysis approach into the EM framework (Cai et al., 2019; Mai et al., 2021), leveraging sparsity assumptions on discriminant factors for dimension reduction. Specifically, these methods require the likelihood ratio $f_1(\mathbf{y}|\boldsymbol{\mu}_1, \boldsymbol{\Sigma})/f_2(\mathbf{y}|\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$ between two component densities to be fully characterized by the difference between two quadratic distances, $(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top (\boldsymbol{\Sigma})^{-1} \{\mathbf{y} - (\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)/2\}$. Nevertheless, the following proposition demonstrates that the discriminant-factor-based methods are applicable only to the mixtures of Gaussian case.

PROPOSITION 1. For an elliptical distribution with density f_k defined in (2.1), its density generator function, $g \in \mathcal{G}$, satisfies that, for any $x_1, x_2 \in \mathbb{R}$, $g(x_1)/g(x_2) = h(x_1 - x_2)$ for some function h , if and only if, the density f_k is from a Gaussian distribution.

2.2. High-resolution feature identification with the PARSE penalty. As discussed earlier, our focus is on achieving finer feature identification by identifying all substantial pairwise differences in mean parameters between clusters while simultaneously clustering. To accomplish this, we introduce a novel penalization-based framework for feature identification under the high-dimensional EM framework. It is important to note that even without a penalty, obtaining the maximum likelihood estimator (MLE) for an arbitrary MoED model in the high-dimensional setting remains challenging due to the complex form of the likelihood function in (2.1). The inclusion of a penalty term further increases computational difficulties. Therefore, with complete data, we consider minimizing the following objective function:

$$(2.3) \quad \tilde{L}_C(\mathbf{Y}, \mathbf{z}, \boldsymbol{\pi}, \mathbf{U}, \boldsymbol{\Sigma}) = n^{-1} \sum_{i=1}^n \sum_{k=1}^K \mathbb{I}(z_i = k) \{(\mathbf{y}_i - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_k) - \log \pi_k\} + \sum_{j=1}^p \sum_{1 \leq k < k'}^K P_{\lambda_n}(|\mu_{kj} - \mu_{k'j}|).$$

Under an EM-type framework, we iterate the clustering estimation (E-step) by calculating the conditional expectation, $\mathbb{E}\{\mathbb{I}(z_i = k)|\mathbf{y}_i, \hat{\boldsymbol{\pi}}, \hat{\mathbf{U}}, \hat{\boldsymbol{\Sigma}}\}$, and the model estimation (M-step) by minimizing $\mathbb{E}_{\mathbf{z}|\mathbf{y}_i, \hat{\boldsymbol{\pi}}, \hat{\mathbf{U}}, \hat{\boldsymbol{\Sigma}}}\{\tilde{L}_C(\mathbf{Y}, \mathbf{z}, \boldsymbol{\pi}, \mathbf{U}, \boldsymbol{\Sigma})\}$. *One key point is that we replace the likelihood in (2.2) by a distance-based quadratic loss $(\mathbf{y}_i - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y}_i - \boldsymbol{\mu}_k)$.* This approach is simple and intuitive under the MoED model, where the component density is a monotonically non-increasing function of such a distance. While the resulting estimator is generally not equivalent to the MLE – except in the Gaussian case – our theoretical analysis demonstrates that the estimator remains efficient and optimal. More importantly, the use of quadratic loss substantially reduces computational burden, particularly when incorporating the additional penalty term. Further discussion is provided in the following section.

For feature identification and sparsity pursuit, a fusion-type penalty $P_{\lambda_n}(\cdot)$ is imposed on pair-cluster-wise differences, for each dimension j . Accordingly, a feature j is identified as (k, k') -**informative** if $\hat{\mu}_{kj} \neq \hat{\mu}_{k'j}$, **overall informative** if at least one pair of clusters exhibits a non-zero difference for that feature, and **non-informative** otherwise. In this framework, the sparsity of signals in the model is significantly more pronounced than in conventional high-dimensional models. Traditional penalties, such as the ℓ_1 -penalization (Tibshirani, 1996) or non-convex penalties like SCAD (Fan and Li, 2001), tend to select overly dense models when signals are extremely sparse (Song and Liang, 2015). This leads to a substantially increased false positive rate in identifying pairwise informative features. To address this challenge, inspired by the reciprocal Lasso (Song and Liang, 2015), we propose the PAirwise Reciprocal fuSE (PARSE) penalty to promote highly sparse models by imposing large penalties on nearly zero differences between clusters. Specifically, the PARSE penalty is defined as

$$(2.4) \quad P_{\lambda_n}(|\mu_{kj} - \mu_{k'j}|) = \lambda_n \mathbb{I}(|\mu_{kj} - \mu_{k'j}| \neq 0) |\mu_{kj} - \mu_{k'j}|^{-1},$$

where the tuning parameter λ_n controls the fusion strength. PARSE profoundly regularizes non-informative features by imposing a penalty that approaches infinity as $|\mu_{kj} - \mu_{k'j}|$ approaches zero. This promotes a high degree of sparsity, improving the identification of pairwise informative features while excluding non-informative ones. In contrast to other options beyond Lasso, such as the adaptive Lasso employed in APFP (Guo et al., 2010), PARSE does not require any pre-estimated weights which usually introduces extra bias for the parameter estimations on informative features.

3. An Adaptive EM Algorithm with Active Feature Selection. Despite offering significant benefits in interpretability and theoretical optimality, optimization for the PARSE model is challenging due to missing cluster labels, high dimensionality, and the imposed non-convex penalty. In this section, we propose a novel EM algorithm framework that is both statistically and computationally efficient.

A critical issue of the high-dimensional EM framework arises in the E-step, where the conditional expectations of latent cluster labels are calculated based on all observed features:

$$(3.1) \quad \hat{\alpha}_{ik} \equiv \mathbb{E}\{\mathbb{I}(z_i = k)|\mathbf{y}_i, \hat{\boldsymbol{\pi}}, \hat{\mathbf{U}}, \hat{\boldsymbol{\Sigma}}\} = \frac{\hat{\pi}_k f_k(\mathbf{y}_i | \hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}})}{\sum_{k'=1}^K \hat{\pi}_{k'} f_{k'}(\mathbf{y}_i | \hat{\boldsymbol{\mu}}_{k'}, \hat{\boldsymbol{\Sigma}})},$$

where we denote $\hat{\mathbf{U}}_{p \times K} = (\hat{\boldsymbol{\mu}}_1, \dots, \hat{\boldsymbol{\mu}}_K)^\top$ similar as the definition of $\mathbf{U}^*$ in Section 1. However, under the sparsity assumption, the cluster structure is only determined by a small subset of informative features, denoted as $\mathcal{J}^*$. Consequently, the optimal Bayes rule for classification depends exclusively on this subset of features, $\mathbf{y}_{i, \mathcal{J}^*}$:

$$(3.2) \quad \alpha_{ik}^* = \mathbb{E}\{\mathbb{I}(z_i = k)|\mathbf{y}_{i, \mathcal{J}^*}, \pi^*, \mathbf{U}^*, \boldsymbol{\Sigma}^*\},$$

where π^* , $\mathbf{U}^*$, and Σ^* are the true parameter values. In the conventional EM algorithm, even though carefully designed regularization may identify non-informative features with homogeneous mean estimates across clusters (i.e., $\hat{\mu}_{kj} = \hat{\mu}_{k'j}$, $1 \leq k \neq k' \leq K$), the estimates of these nuisance parameters will still contribute to the calculation of $\hat{\alpha}_{ik}$. This occurs through the quadratic term, $(\mathbf{y}_i - \hat{\boldsymbol{\mu}}_k)^\top \hat{\Sigma}^{-1}(\mathbf{y}_i - \hat{\boldsymbol{\mu}}_k)$, in the multivariate likelihood $f_k(\mathbf{y}_i | \hat{\boldsymbol{\mu}}_k, \hat{\Sigma})$, unless all features are assumed to be independent and thus the marginal likelihoods with the same parameter estimates are canceled out in (3.1). To bridge this gap, we propose a novel adaptive EM algorithm for high-dimensional clustering that leverages the active selection and incorporation of a tentative set of informative features during the algorithm iterations.

3.1. An adaptive EM framework for PARSE. We consider a pre-specified number of clusters K and a penalization parameter λ_n , with their determinations discussed later.

Adaptive E-step using informative features: At the $(t+1)$ th iteration, given the current model estimates $(\hat{\pi}^{(t)}, \hat{\mathbf{U}}^{(t)}, \hat{\Sigma}^{(t)})$, let $\hat{\mathcal{J}}^{(t)} = \{1 \leq j \leq p : \hat{\mu}_{kj}^{(t)} \neq \hat{\mu}_{k'j}^{(t)} \text{ for some } 1 \leq k < k' \leq K\}$ denote the tentative set of informative features. Then, in the E-step, we calculate the conditional probability of latent indicator z_i only based on the features in $\hat{\mathcal{J}}^{(t)}$:

$$(3.3) \quad \hat{\alpha}_{ik}^{(t+1)} = \frac{\hat{\pi}_k^{(t)} f_k(\mathbf{y}_{i, \hat{\mathcal{J}}^{(t)}} | \hat{\boldsymbol{\mu}}_{k, \hat{\mathcal{J}}^{(t)}}^{(t)}, \hat{\Sigma}_{\hat{\mathcal{J}}^{(t)}, \hat{\mathcal{J}}^{(t)}}^{(t)})}{\sum_{k'=1}^K \hat{\pi}_{k'}^{(t)} f_{k'}(\mathbf{y}_{i, \hat{\mathcal{J}}^{(t)}} | \hat{\boldsymbol{\mu}}_{k', \hat{\mathcal{J}}^{(t)}}^{(t)}, \hat{\Sigma}_{\hat{\mathcal{J}}^{(t)}, \hat{\mathcal{J}}^{(t)}}^{(t)})}$$

where $\mathbf{y}_{i, \hat{\mathcal{J}}^{(t)}}$, $\hat{\boldsymbol{\mu}}_{k, \hat{\mathcal{J}}^{(t)}}^{(t)}$, and $\hat{\Sigma}_{\hat{\mathcal{J}}^{(t)}, \hat{\mathcal{J}}^{(t)}}^{(t)}$ denote the data observation, mean estimates, and covariance matrix estimate on the features in $\hat{\mathcal{J}}^{(t)}$, respectively. Rather than using all features in clustering estimation, focusing on a set of low-dimensional informative features helps mitigate extra estimation errors arising from the aggregation of high-dimensional nuisance parameters, thereby improving the estimation of $\hat{\alpha}_{ik}$. Moreover, this approach also reduces the computational complexity in the E-step by limiting the calculation of component likelihoods in a small subset of features. A similar strategy has also been introduced by Wang et al. (2015), who suggested using only the features with the top $\hat{s}$ largest parameter magnitudes in the ‘‘E’’-step. However, their method cannot be extended to the high-dimensional clustering setting with more than two clusters, and the estimation of tuning parameter $\hat{s}$ is non-trivial.

M-step: Feature-wise optimization with backward selection: Another computational challenge lies in the PARSE penalty for feature identification, which involves the pair-wise fusion and an ℓ_0 -type penalty function. To solve the optimization in the M-step, we employ a ‘‘backward selection’’ procedure with a feature-wise optimization scheme.

Note that the minimization of the quadratic loss in (2.3) does not even rely on the covariance matrix Σ , and the PARSE penalty is imposed on each feature j separately. Therefore, we further simplify the optimization by using an independent working correlation structure in the M-step and then have

$$(3.4) \quad \hat{\mathbf{U}}^{(t+1)} = \arg \min_{\{\mathbf{U}_{\cdot j} = (\mu_{1j}, \dots, \mu_{Kj})^\top\}_{j=1}^p} \sum_{j=1}^p \mathcal{L}_j(\mathbf{U}_{\cdot j} | \hat{\boldsymbol{\theta}}^{(t)}),$$

where

$$(3.5) \quad \mathcal{L}_j(\mathbf{U}_{\cdot j} | \hat{\boldsymbol{\theta}}^{(t)}) = \frac{1}{2n} \sum_{i=1}^n \sum_{k=1}^K \hat{\alpha}_{ik}^{(t+1)} (y_{ij} - \mu_{kj})^2 + \lambda_n \sum_{k < k'}^K \frac{\mathbb{I}(|\mu_{kj} - \mu_{k'j}| \neq 0)}{|\mu_{kj} - \mu_{k'j}|},$$

while the mixing proportion is updated as $\hat{\pi}_k^{(t+1)} = n^{-1} \sum_{i=1}^n \hat{\alpha}_{ik}^{(t+1)}$ for $k = 1, \dots, K$. The feature-wise separability in the above objective function here allows us to naturally adopt a feature-wise optimization scheme, turning the minimization into a number of parallel pieces under the univariate case, which substantially reduces the computational burden and also

enables the effective parallel computing. Our following theoretical analysis also demonstrates that utilizing such an independent working correlation structure will not affect the optimal statistical rate of the obtained estimates.

A backward selection procedure: Owing to the feature-wise separability of the objective function in (3.4), the optimization reduces to p univariate problems. For each t and $j = 1, \dots, p$, the minimizer of $\mathcal{L}_j(\mathbf{U}_{\cdot j} | \hat{\boldsymbol{\theta}}^{(t)})$ falls in a subspace $\bar{M}_j = \{\mathbf{U}_{\cdot j} : \mu_{k_m j} = \mu_{k'_m j}, m = 1, 2, \dots\}$ of $M_{j0} = \{\mathbf{U}_{\cdot j}\}$, where $\mu_{1j}, \dots, \mu_{Kj}$ are all distinct. To solve (3.4), in the first step, let $(k_1, k'_1) = \arg \min_{k \neq k'} \min_{M_{j1}^{k, k'}} \mathcal{L}_j(\mathbf{U}_{\cdot j} | \hat{\boldsymbol{\theta}}^{(t)})$, where $M_{j1}^{k, k'} = \{\mathbf{U}_{\cdot j} : \mu_{kj} = \mu_{k'j}\}$ and denote subspace $M_{j1} = M_{j1}^{k_1, k'_1}$, merging the k_1 th and k'_1 th clusters. Iteratively, we impose additional constraint $\{\mu_{k_b j} = \mu_{k'_b j}\}$ on the subspace of $\mathbf{U}_{\cdot j}$ at each step, and search for target subspace $\bar{M}_j$ satisfying $\{\mu_{k_m j} = \mu_{k'_m j}, m = 1, 2, \dots\}$. Since we obtain a sequence of nested subspaces $M_{j0} \supset M_{j1} \supset \dots$, where $M_{j, b-1} \cap M_{j, b}^c = \{\mathbf{U}_{\cdot j} : \mu_{k_b j} = \mu_{k'_b j}\}$ for $b = 1, 2, \dots$, the search is carried out in a decreasing pattern, moving from M_{j0} to the target subspace $\bar{M}_j$. This procedure successively merges clusters and is similar in spirit to the approaches of Wieczorek and Lei (2022); Shi et al. (2025). It is summarized in Algorithm S.1 in Section C.1 of the supplementary file. Leveraging the feature-wise separability of (3.4), Algorithm S.1 can be shown to converge to the global optimum of (3.4), as detailed in Section C.1. Our subsequent theoretical analysis is therefore carried out directly with respect to the global minimizer of the M-step objective function $\mathcal{L}(\mathbf{U} | \hat{\boldsymbol{\theta}}^{(t)})$ for each t .

ALGORITHM 1 (Alternative Maximization Algorithm for PARSE).

Input: Data $\mathbf{y}_1, \dots, \mathbf{y}_n$, number of clusters K , and tuning parameter λ_n

Output: Cluster assignments $\hat{z}_1, \dots, \hat{z}_n$, parameters $\hat{\mathbf{U}}, \hat{\boldsymbol{\pi}}, \hat{\boldsymbol{\Sigma}}$

Initialize $\hat{\mathbf{U}}^{(0)}, \hat{\boldsymbol{\Sigma}}^{(0)}$, and $\hat{\boldsymbol{\pi}}^{(0)}$.

Do Steps 1a, 1b, 2, 3a, and 3b for $t = 0, 1, 2, \dots$ **until stopping criterion is met, say** $t = T$.

1a. Let $\xi(\hat{\mathbf{U}}^{(t)}) = \{(k, k', j) : \hat{\mu}_{kj}^{(t)} \neq \hat{\mu}_{k'j}^{(t)}, k, k' = 1, \dots, K, j = 1, \dots, p\}$

1b. Let $\hat{\mathcal{J}}^{(t)} = \{j = 1, \dots, p : (k, k', j) \in \xi(\hat{\mathbf{U}}^{(t)}) \text{ for some } k, k' = 1, \dots, K\}$.

2. In the E-step, compute $\hat{\alpha}_{ik}^{(t+1)}$ in (3.3) with $\mathbf{y}_1, \hat{\mathcal{J}}^{(t)}, \dots, \mathbf{y}_n, \hat{\mathcal{J}}^{(t)}, \hat{\pi}_1^{(t)}, \dots, \hat{\pi}_K^{(t)}, \hat{\mu}_{1, \hat{\mathcal{J}}^{(t)}}^{(t)}, \dots, \hat{\mu}_{K, \hat{\mathcal{J}}^{(t)}}^{(t)}$,

and $\hat{\mathcal{J}}^{(t)}, \hat{\mathcal{J}}^{(t)}$:

3a. In the M-step, update $\hat{\pi}_k^{(t+1)} = n^{-1} \sum_{i=1}^n \hat{\alpha}_{ik}^{(t+1)}$ for $k = 1, \dots, K$;

3b. Update $\hat{\mu}_{1, \hat{\mathcal{J}}^{(t)}}^{(t+1)}, \dots, \hat{\mu}_{K, \hat{\mathcal{J}}^{(t)}}^{(t+1)}$ by solving (3.4) using the backward selection procedures as described below;

3c. In the M-step, update $\hat{\boldsymbol{\Sigma}}^{(t+1)}$ as in (3.6);

If stopping criterion $\sum_{k=1}^K \|\hat{\mu}_k^{(t+1)} - \hat{\mu}_k^{(t)}\|_1 \{|\hat{\mu}_k^{(t)}|_1 + \epsilon\}^{-1} + \sum_{j=1}^p |\hat{\sigma}_{jj}^{2, (t+1)} - \hat{\sigma}_{jj}^{2, (t)}| \{|\hat{\sigma}_{jj}^{2, (t)}| + \epsilon\}^{-1} + \sum_{k=1}^K |\hat{\pi}_k^{(t+1)} - \hat{\pi}_k^{(t)}| < \epsilon_0$, **is satisfied for** $\epsilon, \epsilon_0 > 0$, **terminate the loop;**

Compute cluster assignment for each $i = 1, \dots, n$, $\hat{z}_i = \arg \max_k \hat{\alpha}_{ik}^{(T)}$.

Finally, as suggested by the adaptive ‘‘E’’-step, we estimate the covariance matrix only for the informative features in $\hat{\mathcal{J}}^{(t+1)}$ identified in (3.4). This set typically has much smaller dimension $\hat{s}^{(t+1)} = |\hat{\mathcal{J}}^{(t+1)}|$ than p . When $\hat{s}^{(t+1)} < n$, we use a pooled estimator

$$(3.6) \quad \hat{\boldsymbol{\Sigma}}_{\hat{\mathcal{J}}^{(t+1)}, \hat{\mathcal{J}}^{(t+1)}}^{(t+1)} = \frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K \hat{\alpha}_{ik}^{(t+1)} (\mathbf{y}_{i, \hat{\mathcal{J}}^{(t+1)}} - \hat{\mu}_{k, \hat{\mathcal{J}}^{(t+1)}}^{(t+1)}) (\mathbf{y}_{i, \hat{\mathcal{J}}^{(t+1)}} - \hat{\mu}_{k, \hat{\mathcal{J}}^{(t+1)}}^{(t+1)})^\top.$$

If, at some iterations, $\hat{s}^{(t+1)} > n$, that is, the number of tentatively identified informative features exceeds the sample size, making the sample covariance non-invertible, we instead use a diagonal working covariance estimator when computing (3.3). We later show that using this working covariance in certain iterations does not affect the optimal statistical properties of the proposed method. The algorithmic implementation of PARSE is summarized as in

Algorithm 1. In addition to the rigorous theoretical guarantees established later, we provide a few remarks here to offer further insights into the proposed adaptive procedure.

REMARK 1. (a) In high-dimensional model-based clustering, dimension reduction is essential, without which, the true parameter cannot be consistently estimated in the ℓ_2 norm. Thus, the Bayes rule, $\mathbb{E}\{\mathbb{I}(z_i = k) | \mathbf{y}_i, \pi^*, \mathbf{U}^*, \Sigma^*\}$, cannot be consistently recovered through direct estimation of $\pi^*, \mathbf{U}^*, \Sigma^*$ without feature selection. (b) To achieve dimension reduction, we propose an adaptive EM framework that not only reduces the computational complexity of computing component likelihoods, but also bypasses inverting the high-dimensional covariance matrix of all features, $\widehat{\Sigma}$, thereby relaxing the stringent conditions required by existing approaches. The proposed adaptive procedure is not a greedy algorithm: the sequence of informative feature sets, $\widehat{\mathcal{J}}^{(t)}$, is neither nested across iterations nor restricted to a stepwise procedure. The procedure is more essential in a general MoED model rather than the mixture of Gaussian, since even assuming a diagonal covariance structure may fail to eliminate the influence of nuisance parameters on clustering estimations (see supplementary Section C.4).

REMARK 2. In the M-step (3.4), the unpenalized cluster mean estimate $\widehat{\mu}_{kj}$ is simply a weighted average of the sample observations. For simplicity, let us consider a two-cluster case and take a binary cluster indicator $\widehat{\alpha}_{ik}$ for $k = 1, 2$ obtained from the E-step, where $\widehat{\mu}_{1j} = \sum_{i=1}^n \widehat{\alpha}_{i1} y_{ij}$ and $\widehat{\mu}_{2j} = \sum_{i=1}^n \widehat{\alpha}_{i2} y_{ij}$. For a non-informative feature j , the distribution of the difference estimate, $\widehat{\mu}_{1j} - \widehat{\mu}_{2j}$, is largely independent of the specific sample splitting determined by $\widehat{\alpha}_{ik}$. In other words, under random splitting, there is a high probability of screening out a non-informative feature in the M-step by the imposed fusion penalty, even without requiring the estimated weights $\widehat{\alpha}_{ik}$ to closely align with the true cluster labels. Consequently, the proposed algorithm ensures a stable path for dimension reduction, guiding the tentative set $\widehat{\mathcal{J}}^{(t)}$ toward a moderate-sized set containing the informative features.

3.2. *Tuning parameters selection.* To determine the number of clusters K and the tuning parameter λ_n , we use the generalized information criterion (GIC), which is known to consistently identify the true model in high-dimensional settings (Fan and Tang, 2013). We first specify the asymptotic order of the tuning parameter as $\lambda_n = O(n^{3(\gamma-1)/4})$ in our theoretical analysis, which is widely seen in relevant literature (Song and Liang, 2015; Zhao and Yu, 2006; Wainwright, 2009). In practice, we select λ_n and K in a data-driven manner. Specifically, for the PARSE model, let $e(\widehat{\mathbf{U}}_{\cdot j}) \in \{1, \dots, K\}$ denote the number of distinct entries in the j th coordinate, and define $d_{\widehat{\mathbf{U}}} = \sum_{j=1}^p e(\widehat{\mathbf{U}}_{\cdot j})$. Then we calculate $\text{GIC}(\lambda_n, K) = -2L_C(\mathbf{Y}, \widehat{\mathbf{z}}, \widehat{\pi}, \widehat{\mathbf{U}}, \widehat{\Sigma}) + \log\{\log(n)\}c_p$, where $\widehat{z}_i = \arg \max_k \widehat{\alpha}_{ik}$ is the predicted cluster label through the model, L_C is the log-likelihood in (2.2), and $c_p = (K - 1 + p + d_{\widehat{\mathbf{U}}}) \log(K - 1 + p + d_{\widehat{\mathbf{U}}})$. We select λ_n and K by minimizing $\text{GIC}(\lambda_n, K)$ based on a grid search. More empirical studies investigating the selection of these tuning parameters are provided in Section C.2 in the Supplementary File.

In addition, similar to Lasso and its variants (Tibshirani et al., 2005; Zou, 2006), the solution path exhibits a thresholding behavior: there exists a critical value $\lambda_n^{\max}$ such that all estimated pairwise mean differences are zero for $\lambda_n > \lambda_n^{\max}$, in which case all features are identified as non-informative. For further details regarding the derivation of $\lambda_n^{\max}$, please refer to Section C.2 in the supplementary file. In practice, a sequence of candidate values for λ_n is constructed over $(0, \lambda_n^{\max}]$, typically on a logarithmic scale, to facilitate efficient implementation of the algorithm and subsequent model selection.

4. General setting, conditions, and a provable initialization. Before examining the properties of PARSE in Sections 5 and 6, we specify the model parameter space for clarity:

$$(4.1) \quad \Theta = \{(\mathbf{U}, \boldsymbol{\pi}, \boldsymbol{\Sigma}) : \|\mathbf{U}\|_{\infty} \leq C_{\mu}, \quad c_{\pi} \leq \pi_k \leq 1 - c_{\pi}, \quad \sum_{k=1}^K \pi_k = 1\},$$

where $u_{\min} = \min_{(k,k',j) \in \xi(\mathbf{U}^*)} |\mu_{kj} - \mu_{k'j}|$ is the minimum separation between distinct cluster means across all the features, C_μ is a positive constant, and $c_\pi > 0$ ensures a balanced clustering structure, which is common for model-based clustering (Fraley and Raftery, 2002; Cai et al., 2019; Mai et al., 2021; Lyu and Xia, 2022). For $1 \leq k < k' \leq K$, let $s_{kk'} = |\{1 \leq j \leq p : \mu_{kj}^* \neq \mu_{k'j}^*\}|$ be the number of coordinates on which μ_k^* and $\mu_{k'}^*$ differ. Define $s_{\min} = \min_{k < k'} s_{kk'}$ and $s_{\max} = \max_{k < k'} s_{kk'}$. Let $\theta^* = (\mathbf{U}^*, \pi^*, \Sigma^*)$ denote the true parameter values. We impose the following conditions for PARSE.

CONDITION 1. Assume that $\log(p) = O(n^\gamma)$ for some $0 < \gamma < 1$, $s_{\max} = o(\sqrt{n/\log(p)})$, and $s_{\min} u_{\min}^2 \geq C_0$, where C_0 is a constant independent of $n, p, s_{\min}, s_{\max}$ and is specified in Lemma B.3 in the Supplementary File.

CONDITION 2. Assume that $c_\sigma^{-1} \leq \lambda_{\min}(\Sigma^*) \leq \lambda_{\max}(\Sigma^*) \leq c_\sigma$.

CONDITION 3. Given $z_i^* = k$, where $k \in \{1, \dots, K\}$, $\mathbb{E}_{\mathbf{y}}(\mathbf{y}_i - \mu_k^*) = \mathbf{0}$ and $\mathbb{E}_{\mathbf{y}}\{t\mathbf{a}^\top(\mathbf{y}_i - \mu_k^*)\} \leq \exp(t^2 \mathbf{a}^\top \Sigma^* \mathbf{a}/2)$ for any unit vector $\mathbf{a} \in \mathbb{R}^p$ and $|t| \leq 1/\sqrt{\mathbf{a}^\top \Sigma^* \mathbf{a}}$.

Condition 1 controls three aspects of the problem: the allowable growth of p , the number of coordinates that may be informative for separating clusters, and the minimal effective signal required for the hardest-to-separate pair. Allowing $\log(p) = O(n^\gamma)$ with $0 < \gamma < 1$ permits exponentially many features. Comparable sparsity regimes have been studied for Gaussian mixtures (e.g. Yi and Caramanis (2015); Wang et al. (2015)). The requirement $s_{\max} = o(\sqrt{n/\log(p)})$ limits the number of informative coordinates for any given pair of clusters and thus keeps the estimation error of mean differences under control. This scaling is consistent with minimax lower bounds for estimating an $s_{\max}$ -sparse p -dimensional vector (Johnstone, 1994). Moreover, $u_{\min}$ denotes the smallest nonzero between-cluster differences over coordinates, the quantity $s_{\min} u_{\min}^2$ measures the aggregate signal available to separate the weakest pair of clusters, and the constraint $s_{\min} u_{\min}^2 \geq C_0$ ensures this aggregate signal does not vanish. Such a min-signal condition is standard in high-dimensional mixture and discriminant analysis and underlies consistent clustering in procedures that impose sparsity on cluster means or discriminant vectors; see, for example, Cai et al. (2019); Mai et al. (2021). Importantly, when $s_{\min}$ grows with n , the condition on $s_{\min} u_{\min}^2$ allows $u_{\min}$ to decay, provided the aggregate signal for the weakest pair remains bounded away from zero. Since $s_{\min} \leq s_{\max} = o(\sqrt{n/\log(p)})$, we have $u_{\min} \geq O(n^{(\gamma-1)/4})$.

Condition 2 allows a certain degree of dependence among features within each cluster, which is crucial for real-world applications such as scRNA-Seq analysis. Similar assumptions on eigenvalues of covariance matrix have been widely used in the literature (Cai et al., 2019). Condition 3 assumes that the mean-deviation error of the data follows a sub-exponential distribution, which is typical in a high-dimensional setting. Together with the consideration on MoED models, the proposal approach covers a reasonable family of commonly used distributions. For example, Condition 3 is satisfied if $\mathbf{y}_1$ follows a multivariate Gaussian with $g(x) = C \exp(-x/2)$, an elliptically symmetric logistic distribution with $g(x) = C \exp(-x)/\{1 + \exp(-x)\}$, a multivariate Pearson type II distribution with $g(x) = C(1-x)^{\beta/2-1}$ with $\beta > 0$, and a double exponential distribution with $g(x) = C \exp(-\sqrt{2}x)$. Compare to the commonly used Gaussian mixtures (Witten and Tibshirani, 2012; Jin and Wang, 2016), Condition 3 substantially relaxes the distributional assumptions of the data.

For implementing the EM algorithm and its variants, a satisfactory initialization is indispensable. When $K = 2$, where the cluster means are symmetric up to a location transformation, the EM algorithm converges with random initialization (Xu et al., 2016; Wu and Zhou,

2021), provided the initial cluster mean is not orthogonal to the true mean μ_1^* . However, random initialization can lead to pathological fixed points for three-component Gaussian mixtures (Jin et al., 2016). To ensure achieving the optimal theoretical convergence, we impose the following initialization condition and develop an initialization procedure that satisfies this condition with provable guarantees.

CONDITION 4. Assume an initialization $\hat{\mathbf{U}}^{(0)}$ exists such that, for some permutation Π of $\{1, \dots, K\}$, $\max_j |\hat{\mu}_{kj}^{(0)} - \mu_{\Pi kj}^*| \leq u_{\min}/2$ for each $k = 1, \dots, K$.

4.1. *A spectral-based initialization.* Motivated by recent advances in spectral clustering for mixture models (Zhang and Zhou, 2024), we propose a spectral-clustering-based initialization for PARSE that is easy to implement and computationally efficient. The implementation details are summarized in Algorithm S.2 in Section C.3 of the supplementary file. In the following proposition, we show that this proposed initialization satisfies Condition 4.

PROPOSITION 2. Under Conditions 1-3, further assuming $s_{\min}^{1/2} u_{\min}^2 \geq C'_0$, where C'_0 is a positive constant, for a sufficiently large n , the obtained estimate $\hat{\mathbf{U}}^{(0)}$ from Algorithm S.2 satisfies that with probability at least $1 - n^{-1} - \exp(-n^{(\gamma+1)/2})$, $\max_j |\hat{\mu}_{kj}^{(0)} - \mu_{\Pi kj}^*| \leq u_{\min}/2$ for each $k = 1, \dots, K$, where Π is a permutation of $\{1, \dots, K\}$.

Note that Proposition 2 imposes a slightly stronger condition on $u_{\min}$ to ensure that the initialization obtained from Algorithm S.2 satisfies the requirement in Condition 4. This arises because Algorithm S.2, as a spectral-clustering-based procedure, does not leverage sparsity in informative clustering features. The $u_{\min}$ -condition in Proposition 2 is equivalent to $s_{\min} u_{\min}^2 \geq C'_0 s_{\min}^{1/2}$. This, to some extent, aligns with the standard results for conventional high-dimensional spectral-clustering algorithms (Fan et al., 2018; Löffler et al., 2021; Zhang and Zhou, 2024; Huang and Gu, 2025), requiring $s_{\min} u_{\min}^2$ to diverge with the growing $s_{\min}$.

Next, we further clarify the relationships between general conditions (Conditions 1 and 4) and those in Proposition 2. Condition 1 specifies the general regime for data structure, including the sample size n , overall feature dimension p , sparsity levels (signal dimension) $s_{\min}$ and $s_{\max}$, and the minimal signal strength $u_{\min}$. These conditions are standard in the high-dimensional regime in order to establish the statistical properties of the global-optimum estimators. However, due to the nonconvexity of the clustering objective, convergence to the global optimum is not guaranteed in general. Therefore, our theoretical analysis is carried out with respect to the EM iterates under a suitable initialization. Condition 4 formalizes this requirement of a reasonable initialization, which is a common assumption in the analysis of EM-type algorithms (e.g., Balakrishnan et al., 2017; Hao et al., 2017; Cai et al., 2019). Conditions 1 and 4 provide an overview and theoretical boundary for the proposed PARSE framework, without relying on any specific initialization algorithms. In practical applications, we propose the use of a spectral-clustering-based algorithm (Algorithm S.2) for initialization, which is easy to implement and computationally stable. Proposition 2 justifies the validity of Algorithm S.2, showing that it satisfies Condition 4 under mild conditions.

Notably, several other initialization methods can also be used and will satisfy Condition 4 under weaker conditions for specific scenarios. The HP algorithm (Hardt and Price, 2015), based on the method of moments, applies only to $K = 2$, and to our knowledge, satisfies Condition 4 without additional signal-strength assumptions only when $K = 2$. The initialization via ordered distance (IOD) algorithm (Jana et al., 2024) constructs initializations using ordered distances between cluster means and may serve as an alternative to Algorithm S.2. For general K , moderate p , and non-vanishing $u_{\min}$, the IOD algorithm requires a condition

similar to Condition 1, namely that $s_{\min} u_{\min}^2$ exceeds a constant, and ensures Condition 4. However, the IOD algorithm requires p to be finite and $u_{\min}$ to be non-vanishing, limiting its applicability for our proposed MoED model.

5. Statistical Guarantees of PARSE on the informative feature identification. A key focus of PARSE is to simultaneously identify interpretable informative features and perform clustering. In this section, we first discuss the theoretical strategy for studying PARSE, followed by establishing its consistency and optimality in feature identification.

5.1. Strategy to study PARSE. The statistical guarantees of PARSE implemented by Algorithm 1 follow from the convergence of the proposed adaptive EM algorithm, established by comparing the empirical updates with their population counterparts. The optimization in (3.4) is central for analyzing statistical errors, with the posterior probabilities $\hat{\alpha}_{ik}^{(t)}$ from the adaptive “E” step playing a key role. In our theoretical analysis, we link the sample loss function in (3.4) and the corresponding minimizer $\hat{\mathbf{U}}^{(t)}$ to its population variant and the corresponding minimizer $\tilde{\mathbf{U}}^{(t)}$ (see the definition in (SA.2) in the Supplementary File). Essentially, the population path $\{\tilde{\mathbf{U}}^{(t)}\}_{t=1,2,\dots}$, is used to characterize the contraction of the EM iterates at the population level, while the deviation between $\hat{\mathbf{U}}^{(t)}$ and $\tilde{\mathbf{U}}^{(t)}$ captures the statistical error arising from sample fluctuations. By combining two solution paths, we decompose the overall estimation error of the PARSE estimator into an algorithmic component and a statistical component, which leads to the main results in Theorem 1. Specifically, in Lemmas B.5 to B.7 in Section B in the Supplementary File, we show that $\hat{\mathbf{U}}^{(t)}$ is uniformly concentrated around $\tilde{\mathbf{U}}^{(t)}$ across iterates. Then, for sufficiently large t , the computational error will be dominated by the statistical error.

A key challenge in our analysis is the set of features used in the adaptive “E”-step is updated across iterations, thereby making it nontrivial to align the sequence of sample loss functions with their population counterparts. To overcome this difficulty, we introduce an intermediate path defined through a population-level loss while retaining the sample-based posterior updates, which allows us to bridge the sample and population paths. Moreover, the singularity and non-smooth nature of the reciprocal penalty in PARSE creates additional difficulties in controlling the difference between different paths. To circumvent this, we develop a new argument that goes beyond conventional approaches: in addition to controlling the ℓ_2 -distance between different paths, we establish alignment of the corresponding feature-selection sets along them. See Lemma B.7 in the Supplementary File for details.

5.2. Consistency of PARSE on the informative feature identification. In the following theorem, we first characterize the finite sample errors in recovering $\mathbf{U}$ using PARSE, which then leads to the consistency in feature identification.

THEOREM 1. *Assume Conditions 1–4 hold. Let $\lambda_n = O(n^{3(\gamma-1)/4})$. The proposed PARSE estimates implemented via Algorithm 1 satisfy:*

(a) *With a sufficiently large n , for each iteration t , with probability at least $1 - 3\exp(-n^{(\gamma+3)/8})$,*

$$\|\hat{\mathbf{U}}^{(t)} - \mathbf{U}^*\|_{\infty} \leq \kappa^t \|\hat{\mathbf{U}}^{(0)} - \mathbf{U}^*\|_{\infty} + r_n,$$

where $0 < \kappa < 1$ and $r_n = O(\sqrt{s_{\max} \log(p)/n})$. Furthermore, for $T_0 \geq \{\log(\kappa)\}^{-1} \log(\|\hat{\mathbf{U}}^{(0)} - \mathbf{U}^\|_{\infty}/r_n)$,*

$$\|\hat{\mathbf{U}}^{(T_0)} - \mathbf{U}^*\|_{\infty} \leq 2r_n.$$

(b) For a sufficiently large n ,

$$(5.1) \quad \mathbb{P}\{\xi(\widehat{\mathbf{U}}^{(T_0)}) \neq \xi(\mathbf{U}^*)\} \leq 3 \exp(-n^{(\gamma+3)/8}).$$

Theorem 1 characterizes the iterated estimates from the PARSE algorithm. Specifically, the error decomposition demonstrated in Theorem 1(a) indicates that the algorithmic error $\kappa^t \|\widehat{\mathbf{U}}^{(0)} - \mathbf{U}^*\|_\infty$ decreases iteratively and will eventually be dominated by the statistical error r_n . Theorem 1(b) provides the selection consistency in identifying both the overall and the pairwise informative features. In (5.1), the probability of $3 \exp(-n^{(\gamma+3)/8})$ is derived from the separation level between two component distributions based on the scale of $u_{\min}$. The requirement on the tuning parameter λ_n is standard in high-dimensional analysis, and its practical selection is discussed in detail in Section 3.2. Essentially, the order of λ_n is associated with the feature-wise signal strength $u_{\min}$. To avoid spurious merges under the imposed ℓ_0 -penalty, we therefore require the coefficient of fusion penalty, which is bounded by $\lambda_n/u_{\min}^2$, to be smaller than the minimum signal $u_{\min}$. Hence, we ask for $\lambda_n \lesssim u_{\min}^3$, resulting in the required rate for λ_n .

5.3. Optimality of PARSE on the informative feature identification. As noted in Section 1.1, within the model-based clustering framework, consistency and convergence rates for estimating cluster means have been well studied, while the fundamental difficulty of feature identification remains underexplored. Moreover, multiple testing approaches for identifying informative features often require accurate clustering beforehand yet primarily focus on consistency, without addressing optimality. Below, we establish the lower bound for identifying informative features to study the optimality of PARSE on this task.

THEOREM 2. *Under Conditions 1-3, for some positive $\eta = o(1)$, the minimax lower bound of informative feature identification risk satisfies*

$$(5.2) \quad \inf_{\widehat{\mathbf{U}}_n} \sup_{\theta \in \Theta, g \in \mathcal{G}} \mathbb{P}(\xi(\widehat{\mathbf{U}}_n) \neq \xi(\mathbf{U})) \geq \eta.$$

To establish Theorem 2, we construct $\Theta_1 \subset \Theta$ such that, for $(\mathbf{U}, 1/K, \dots, 1/K, \mathbf{I}) \in \Theta_1$ and each $j \in \xi(\mathbf{U})$, exactly $K - 1$ clusters share the same mean, leaving one cluster with a distinct mean differing by at least $u_{\min}$. We further design a subspace $\Theta_2 \subset \Theta_1$ with $|\Theta_2| = p \cdot s_{\max}$, to serve as a desired covering of Θ so that the standard Fano's method applies for deriving the lower bound for feature identification errors. Together with Theorem 1(b), Theorem 2 shows that PARSE is optimal for informative feature identification within the parameter space Θ under the MoED model.

As discussed in Section 2.2, PARSE performs high-resolution feature identification by recovering the set of triplets $\xi(\mathbf{U})$ that index all nonzero pairwise differences among cluster means. A coarser, low-resolution alternative would instead recover only the set of informative coordinates $\mathcal{J} = \{j : (k, k', j) \in \xi(\mathbf{U}) \text{ for some } 1 \leq k < k' \leq K\}$, that is, the features on which at least one pair of clusters differs. To characterize the signal strength required for this task, define $v_{\min} = \min_{j \in \mathcal{J}} \max_{\mu_{kj} \neq \mu_{k'j}} |\mu_{kj} - \mu_{k'j}|$, which satisfies $v_{\min} \geq u_{\min}$, since $u_{\min} = \min_{j \in \mathcal{J}} \min_{\mu_{kj} \neq \mu_{k'j}} |\mu_{kj} - \mu_{k'j}|$. The distinction reflects the different resolutions of the identification tasks. For high-resolution recovery of $\xi(\mathbf{U})$, one must detect every nonzero pairwise difference, so the difficulty is governed by the smallest such signal $u_{\min}$. In contrast, identifying $\mathcal{J}$ only requires detecting at least one pairwise difference for each informative feature j , and the relevant signal strength is therefore $v_{\min}$, the largest separation among cluster pairs at that coordinate. Consequently, $v_{\min}$ quantifies the difficulty of low-resolution feature identification under the MoED model, while $u_{\min}$ characterizes the stronger signal requirement needed for the high-resolution recovery performed by PARSE.

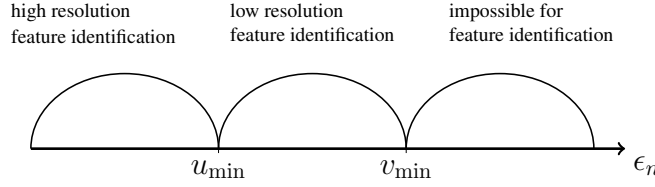


Fig 2: Transition of the difficulties in low and high-resolution feature identification when ϵ_n varies across different regions.

Recall that in Condition 1, the signal strength is required to exceed the error rate $\epsilon_n = O(n^{(\gamma-1)/4})$ for $0 < \gamma < 1$ to achieve consistent and oracle feature identification. Since $u_{\min} \leq v_{\min}$, the relative rates of $u_{\min}$ and $v_{\min}$ to ϵ_n define three distinct cases. First, if Condition 1 holds and $u_{\min} \geq \epsilon_n$, Theorems 2 ensures that high-resolution feature identification is possible. If $u_{\min} < \epsilon_n \leq v_{\min}$, then $\inf_{\hat{U}_n} \sup_{\theta \in \Theta, g \in \mathcal{G}} \mathbb{P}(\xi(\hat{U}_n) \neq \xi(U)) \geq O(1)$ and $\inf_{\hat{U}_n} \sup_{\theta \in \Theta, g \in \mathcal{G}} \mathbb{P}(\hat{\mathcal{J}} \neq \mathcal{J}) \geq o(1)$, for which only low-resolution feature identification is achievable. Finally, if $v_{\min} < \epsilon_n$, even low-resolution feature identification becomes impossible. These are summarized in Figure 2.

6. Consistency and Optimality of PARSE on Clustering. To study the guarantees on clustering accuracy of clustering rule $\hat{\delta}$ produced by PARSE, we consider the permutation invariant mis-clustering error $\mathcal{R}(\hat{\mathbf{z}}) = \min_{\Pi \in \mathcal{S}_K} \mathbb{E}[\mathbb{I}\{\Pi\hat{\delta}(\mathbf{y}_{\text{new}}) \neq z_{\text{new}}^*\}]$, where $\mathcal{S}_K$ is the set of permutations of $\{1, \dots, K\}$ and z_{new}^* is the true label of future observation $\mathbf{y}_{\text{new}}$. Let $\hat{\mathbf{z}}^{\text{opt}}$ denote the estimate labels from the optimal Bayes rule using the true informative set $\mathcal{J}^*$, where $\hat{\mathbf{z}}_{\text{new}}^{\text{opt}} = \arg \max_k \pi_k^* f_{k, \mathcal{J}^*}(\mathbf{y}_{\text{new}, \mathcal{J}^*} | \boldsymbol{\mu}_{k, \mathcal{J}^*}^*, \boldsymbol{\Sigma}_{\mathcal{J}^*}^*)$.

6.1. Fundamental limit on clustering under the MoED model. To establish the fundamental difficulty of clustering under the MoED model, we examine the difference in mis-clustering errors between any clustering rule and the optimal Bayes rule, and the minimax lower bound of the difference as formalized below.

THEOREM 3. *Under Conditions 1-3, MoED model with parameter space in (4.1) satisfies that*

$$\inf_{\hat{\delta}} \sup_{\theta \in \Theta, g \in \mathcal{G}} \mathbb{E}_{\mathbf{y}} \{ \mathcal{R}(\hat{\delta}(\theta, g)) - \mathcal{R}(\hat{\mathbf{z}}^{\text{opt}}(\theta, g)) \} \gtrsim s_{\max} \log(p) n^{-1},$$

where the infimum is taken over all clustering rules $\hat{\delta}: \mathbf{y}_i \rightarrow \{1, \dots, K\}$ for $i = 1, \dots, n$.

The MoED model with the parameter space in (4.1) includes the Gaussian mixture model as a special case, for which similar minimax lower bound is obtained in Cai et al. (2019). From Theorem 3, since the number of informative features for clustering is of order $s_{\max}$, the gap between the best clustering rule and the optimal Bayes rule scales as $s_{\max} \log(p)/n$ and converges to zero as n increases. We emphasize again that the difference in mis-clustering errors between any clustering rule and the optimal Bayes rule is crucial here. Assuming sub-Gaussian component distributions, the minimax lower bound for $\mathbb{E}_{\mathbf{y}} \{ \mathcal{R}(\hat{\delta}) \}$ is shown to be $\exp(-(1 + C_1)s_{\min} u_{\min}^2 c_{\sigma}^{-1}/8)$ when $u_{\min}$ exceeds a constant, for some constant $C_1 > 0$ (Lyu and Xia, 2022; Zhang and Zhou, 2024). This matches the error rate of the optimal Bayes rule. It is evident that $s_{\min}$ or $u_{\min}$ must diverge for the lower bound to converge to zero. By contrast, we focus on the challenging case of potentially shrinking signal strength, as specified by $u_{\min}$ in Condition 1. In this setting, even the accuracy of the optimal Bayes rule

may approach zero, making comparisons between a clustering rule and the optimal Bayes rule the only meaningful benchmark. That is, on this broader parameter space with relaxed signal strength, we aim to recover the optimal Bayes rule $\hat{\mathbf{z}}^{\text{opt}}$ rather than the true clustering labels $\mathbf{z}^*$. When $u_{\min}$ diverges, the optimal Bayes rule converges to the true clustering labels, making clustering less challenging. In this case, the optimal method achieves perfect recovery of clusters, and benchmarking against the optimal Bayes rule effectively becomes benchmarking against the true labels.

6.2. Clustering optimality via PARSE. In the following theorem, we establish the upper bound on the clustering error for PARSE using our adaptive EM algorithm proposed in Section 3, demonstrating its desired optimality.

THEOREM 4. *Under conditions of Theorem 1, with probability at least $1 - 3\exp(-n^{(\gamma+3)/8})$, PARSE implemented via Algorithm 1 satisfies that*

$$\mathcal{R}(\hat{\mathbf{z}}) - \mathcal{R}(\hat{\mathbf{z}}^{\text{opt}}) \lesssim s_{\max} \log(p) n^{-1}.$$

For Gaussian mixture model without extra structural assumptions, Hao et al. (2017) showed that the squared error for estimating cluster means and covariances is of $O_P(p/n)$, leading to that $\mathcal{R}(\hat{\mathbf{z}}) - \mathcal{R}(\hat{\mathbf{z}}^{\text{opt}}) = O_P(p/n)$, which converge to zero unless $p = o(n)$. By incorporating $s_{\max}$ -sparse cluster means or discriminant vectors into Gaussian mixtures, the squared error in estimating cluster means is of the order is $O_P(s_{\max} \log(p)/n)$, resulting in $\mathcal{R}(\hat{\mathbf{z}}) - \mathcal{R}(\hat{\mathbf{z}}^{\text{opt}}) = O_P(s_{\max} \log(p)/n)$ under certain regularity conditions (Yi and Caramanis, 2015; Wang et al., 2015; Cai et al., 2019).

We highlight the distinctions between the seemingly identical clustering error rates of PARSE and other methods. First, the sparsity in Theorem 4 differs from that in other model-based clustering methods that rely on discriminant vectors. Specifically, PARSE imposes sparsity on cluster means to identify interpretable informative features, whereas other methods impose sparsity on affine transformations of all features. Thus, the benchmark Bayes optimal rule for PARSE also differs from that of other methods. Second, a key step to derive results similar to Theorem 4 for high dimensional model-based clustering is analyzing the posterior proportions of clusters using the likelihood ratio based on estimated mixture parameters. This typically relies on Gaussianity, along with either additional assumptions on weak correlations among all features or the use of discriminant vectors instead of raw features, ensuring the likelihood ratio is computable or reasonably approximated, as noted in Section 2.1. Without Gaussianity, however, uncorrelated features do not necessarily imply independence, which prevents the conventional analysis of the likelihood or its approximation. See Section C.4 in the supplementary file for more discussions. Moreover, as Proposition 1 shows, the discriminant vector approach becomes inapplicable under the MoED model. In contrast, the adaptive E-step and PARSE penalty in M-step not only provide efficient computational advantages but also enable our theoretical analysis of clustering errors of PARSE under the MoED model.

Together with Theorems 3, it shows that PARSE, combined with the proposed adaptive EM algorithm, is optimal for clustering with interpretable informative features. In addition, without assuming sparsity, Theorem 4 implies that the clustering error rate is $O(p \log(p)/n)$, which is nearly optimal for clustering without structural assumption, up to a logarithm factor.

7. Simulations. In this section, we evaluate the performance of the proposed PARSE method for clustering and identifying clustering-relevant features. For clustering accuracy, PARSE is compared with competitors including the APFP, CHIME, *K-means*, sparse *K-means* (SK-means, Witten and Tibshirani (2012)), sparse convex clustering (SCVX, Wang

et al. (2018)), and IF-PCA (Jin and Wang, 2016). For identifying cluster-relevant features, we consider APFP, CHIME, and a two-stage procedure as competing methods. The two-stage procedure first applies *K-means* (“2S-Kmeans”) or sparse *K-means* (“2S-SKmeans”) to obtain clustering labels, followed by feature-wise ANOVA with estimated clusters as groups to identify features with p -values below $0.05/p$. This approach mirrors the widely used post-clustering pipeline in genomics (Zhang et al., 2019; Ma and Zhou, 2024). The parameter λ_n for PARSE is tuned as described in Section 3.2, while the competing methods use default tuning parameters, with the true number of clusters K provided.

All code is run on a laptop with a 2.6 GHz Intel Core i7 processor and 16 GB of memory. The computation times of PARSE and competing methods under Scenario 1 are reported in Figure S.1 of the supplementary file, showing that PARSE has comparable computational cost to the competing methods.

7.1. Settings. Data are generated under three scenarios with different distributions and feature dependencies, as well as varying dimensions p , sample sizes n , sparsity s , and signal-noise-ratio ω . Each scenario is constructed with $K = 4$ balanced clusters.

Scenario 1 Data are generated from a p -dimensional Gaussian mixture $\sum_{k=1}^4 0.25\mathcal{N}(\mu_k, \Sigma)$, where Σ is the identity matrix. The cluster means are defined as follows: $\mu_1 = (2.5\omega, \dots, 2.5\omega, 1.5\omega, \dots, 1.5\omega, 0, \dots, 0)^\top$, with the first $s/2$ entries as 2.5ω , the next $s/2$ as 1.5ω , and the rest as 0; $\mu_2 = (0, \dots, 0, 1.5\omega, \dots, 1.5\omega, 0, \dots, 0)^\top$, with the $(s/2 + 1)$ th to s th entries as 1.5ω and the rest as 0; $\mu_3 = (0, \dots, 0, -1.5\omega, \dots, -1.5\omega, 0, \dots, 0)^\top$, with the $(s/2 + 1)$ th to s th entries as -1.5ω and the rest as 0; and $\mu_4 = (-2.5\omega, \dots, -2.5\omega, -1.5\omega, \dots, -1.5\omega, 0, \dots, 0)^\top$, with the first $s/2$ entries as -2.5ω , the next $s/2$ th as -1.5ω , and the rest as 0. Thus, the first $s/2$ features are informative for separating clusters 1 and 4, the next $s/2$ features are informative for separating clusters 1-2 from clusters 3-4, and the remaining features are non-informative for clustering.

Scenario 2 Data are generated from a p -dimensional mixture $\sum_{k=1}^4 0.25f_k(\mu_k, \Sigma)$, with cluster means μ_k ’s as in **Scenario 1**. Each f_k is a non-central multivariate symmetric variance-gamma distribution with mean μ_k , covariance Σ , and density $f_k(\mathbf{y}; \mu_k, \Sigma) = 2\{(2\pi)^{p/2}\Gamma(p/2)|\Sigma|^{1/2}\}^{-1}\mathcal{K}_0(2\sqrt{(\mathbf{y} - \mu_k)^\top \Sigma^{-1}(\mathbf{y} - \mu_k)/p})$, with $\mathcal{K}_0(x) = \int_0^\infty \exp\{-x \cosh(t)\}dt$. The covariance Σ is taken as the sample correlation matrix of the first p dimensions of the human pancreatic cells data in Section 8.

Scenario 3 Data are generated as in **Scenario 1**, except that the covariance matrix is $\Sigma = \{\sigma_{ij}\}_{i,j=1}^p$, where $\sigma_{ii} = 1$ and $\sigma_{ij} = 0.8^{|i-j|}$ for $i \neq j$.

For each scenario, we vary n from 20 to 200, p from 100 to 5,000, s from 20 to 60, and ω from 0.25 to 2.5. All results are based on 100 replicates. Since SCVX is computationally intensive and can be unstable for large p , its results for $p = 5,000$ are omitted.

7.2. Results on clustering accuracy. To evaluate clustering performance, we use the empirical version of $\mathcal{R}(\hat{\mathbf{z}})$: $\hat{\mathcal{R}}(\hat{\mathbf{z}}) = \min_{\Pi \in \mathcal{S}_K} n^{-1} \sum_{i=1}^n \mathbb{I}\{\Pi(\hat{z}_i) \neq z_i^*\}$. As shown in Figure 3, PARSE outperforms all competing methods across Scenarios 1-3. In Scenario 2, where feature dependencies in the synthetic data closely mimic those in real data, such dependence slightly degrades PARSE’s performance; nevertheless, it remains the best-performing method. This robustness is expected, since PARSE requires only sub-exponential errors, whereas CHIME and IF-PCA rely on Gaussian or sub-Gaussian assumptions and therefore deteriorate under dependence. In Scenario 3, where feature dependence follows an AR(1) model, PARSE’s convergence is unaffected. These results demonstrate that Condition 2, as discussed in Section 4, is fairly weak, making PARSE broadly applicable, particularly to genomics and related data.

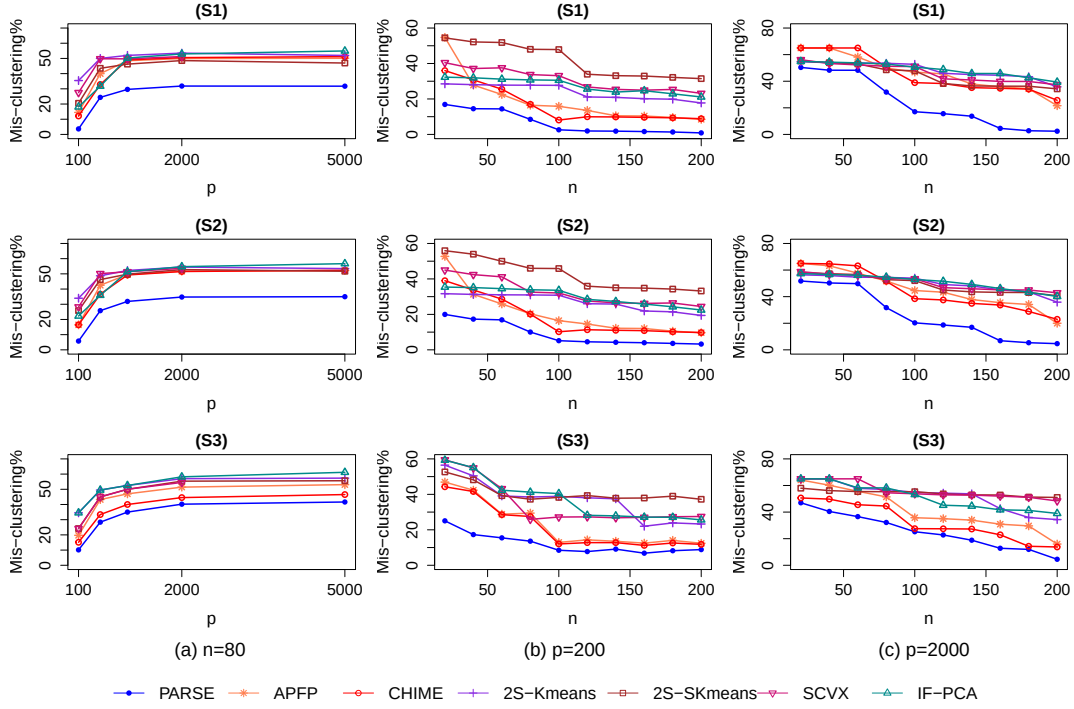


Fig 3: Mis-clustering error (Mis-Clustering%) (a) when $n = 80$, $s = 20$, $\omega = 1$, and p varies from 100 to 5,000; (b) when $p = 200$, $s = 20$, $\omega = 1$, and n varies from 20 to 200; and (c) when $p = 2,000$, $s = 20$, $\omega = 1$, and n varies from 20 to 200. Rows 1 (S1) to 3 (S3) correspond to Scenarios 1 to 3, respectively.

7.3. Results on feature identification. To evaluate feature identification performance, we divide the features into three groups: the first $s/2$ features distinguish clusters 1,4 from 2,3; the next $s/2$ features distinguish clusters 1,2 from 3,4; and the remaining $p - s$ features are irrelevant for clustering. Performance on identifying these three groups assesses both the accuracy of estimating $\xi(\mathbf{U}^*)$ and the ability of high-resolution feature identification. For each method, we compute the metrics $\text{Info}\% = |\{\text{truly informative features}\} \cap \{\text{informative features detected by PARSE}\}| / |\{\text{truly informative features}\}|$, and $\text{NonInfo}\% = |\{\text{truly non-informative features}\} \cap \{\text{informative features detected by PARSE}\}| / |\{\text{truly non-informative features}\}|$, separately for the first two feature groups. $\text{Info}\%$ is the proportion of truly informative features correctly identified as informative and therefore reflects the power of feature selection. In contrast, $\text{NonInfo}\%$ is the proportion of truly non-informative features incorrectly flagged as informative, corresponding to the false positive rate.

Figure 4 shows that as the dimension p increases while the number of informative features and the signal-to-noise ratio remain fixed, the performance of all methods declines. However, PARSE consistently outperforms other methods across all settings. As p increases and the number of informative features remains fixed, the number of non-informative features grows, leading to an expected decrease in $\text{NonInfo}\%$. In Scenario 2, where feature dependencies are more complex as derived from the real data, PARSE achieves a greater performance margin compared to other methods for both $\text{Info}\%$ and $\text{NonInfo}\%$, as shown in Figure 4(S2-a)-(S2-d). In Scenario 3, shown in Figure 4(S3-a)-(S3-d), the dependence between informative and non-informative features make it more challenging to correctly identify informative features. Hence, compared to Scenario 1 in Figure 4(S1-a)-(S1-d), $\text{Info}\%$ decreases while $\text{NonInfo}\%$ increases for all methods. Nevertheless, PARSE remains the least affected among all meth-

ods. Also, the second feature group exhibits higher recovery accuracy, as reflected by larger

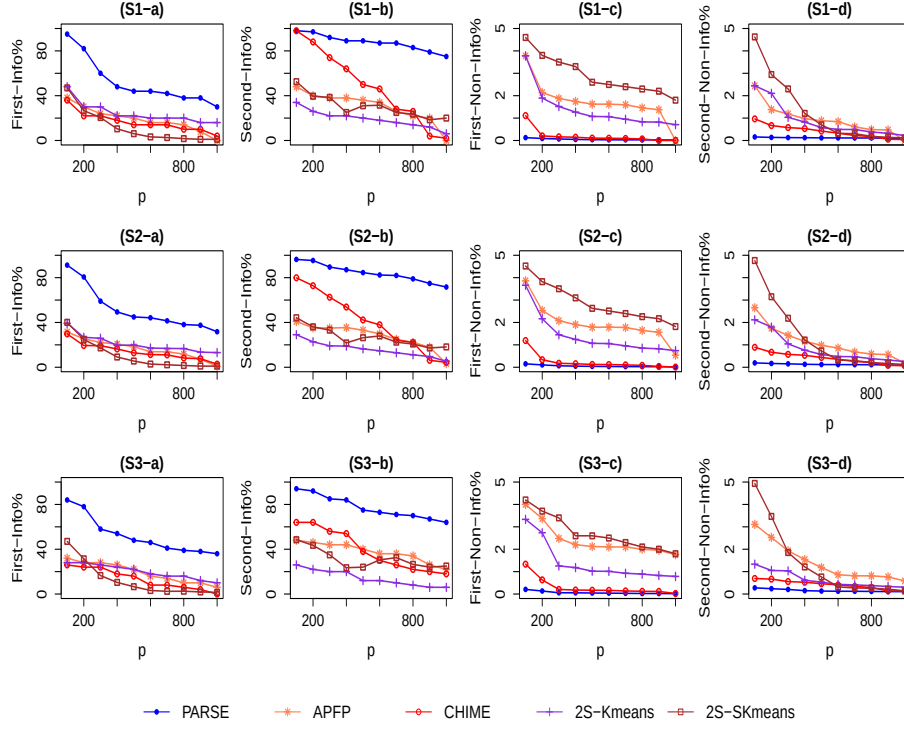


Fig 4: Columns (from left to right): First-Info% (Info% for the first feature group), Second-Info% (Info% for the second feature group), First-NonInfo% (NonInfo% for the first feature group), and Second-NonInfo% (NonInfo% for the second feature group). Rows 1 (S1)-3 (S3) correspond to Scenarios 1-3, respectively. Results are on $n = 80$, $s = 20$, $\omega = 1$, with p varying from 100 to 1,000.

Second-Info% than First-Info%, particularly when p is large. This is because the informative features in the second group induce a simpler two-cluster structure, whereas those in the first group induce a more complex three-cluster structure. Consequently, the informative features in the second group are easier to identify, especially under high dimensionality or complex covariance structures (Scenarios 2-3).

Since the differences among cluster means are sparse, SK-means outperforms *K-means*, as shown in Figure 3. Thus, 2S-SKmeans achieves a lower NonInfo%. However, because strengths of the first $s/2$ features separating clusters 1 and 4 are stronger than those of the second $s/2$ features separating clusters 1-2 from clusters 3-4, SK-means tends to use the first $s/2$ features while overlooking the second $s/2$ features. As a result, Info% for the second group in 2S-SKmeans is lower than that of 2S-Kmeans.

Additional results in Figures S.4 to S.6 in the supplementary file show that all methods improve with increasing sample size and signal-noise-ratio. However, when only sparsity s increases, PARSE continues to deliver the best Info% and NonInfo%, while other methods fail to leverage the increased sparsity, as shown in Figure S.6.

8. Identification of Cell Populations and Relevant Genes in Human Pancreatic Cells.

To demonstrate PARSE's performance in practice, we apply it to human islet cells (HIC) data

(Xin et al., 2016)¹, which contain expression levels of 39,851 genes across 1,600 human pancreatic cells. The cells include glucagon-producing α -cells (glucagon, a hyperglycemic hormone), insulin-producing β -cells (lower blood glucose levels), somatostatin-producing δ -cells (maintain glucose homeostasis), and pancreatic polypeptide (PP) cells (regulate insulin and glucagon secretion) (Röder et al., 2016). Of the 39,851 genes, 5,962 have zero expression across all samples and are excluded from the analysis. For the remaining data, we apply a $\log(y + 1)$ transformation to each observation, as recommended in the literature (Chung et al., 2021); a detailed discussion is provided in Section E of the supplementary file.

8.1. Cell subpopulation identification using PARSE. For cell subpopulation identification with PARSE, we consider $K = 3, 4, 5$ and select $K = 4$ based on $\text{GIC}(\lambda_n, K)$. Among the 33,889 genes, PARSE identifies 839 as informative, referred to as PARSE-genes, which are further divided into six groups to distinguish the four cell subpopulations. The number of PARSE-genes for each pair of cell subpopulations is shown in Figure 5. For clustering performance, we compare PARSE with APFP, CHIME, K -means, SK -means, Seurat (Hao et al., 2021), and SINCERA (Guo et al., 2015). APFP, CHIME, K -means, and SK -means are applied using the true number of clusters, while Seurat and SINCERA are tuned to produce four clusters. Given the large number of genes, we evaluate competing methods using both all genes and PARSE-genes. Using PARSE-genes substantially improves clustering accuracy across all methods, demonstrating their strong informativeness for cell subpopulation identification. Notably, the widely used Seurat pipeline, when combined with PARSE-genes, shows substantial improvement, providing clear evidence of PARSE’s practical superiority.

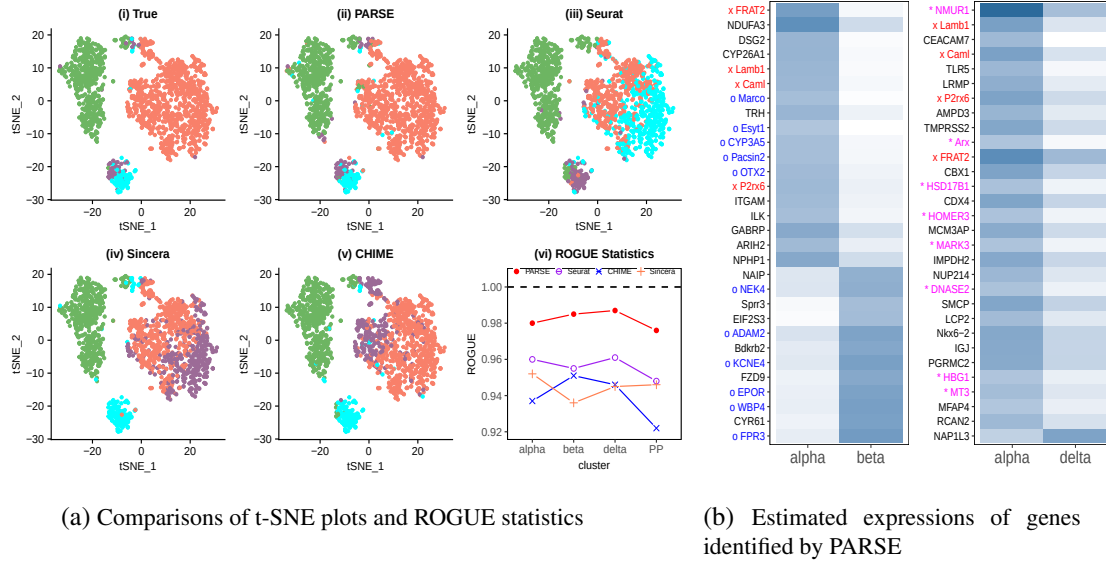
TABLE 1
Clustering accuracy (%) to identify cell subpopulations for the HIC data.

Dataset	PARSE	APFP	CHIME	K -means	SK -means	SEURAT	SINCERA
All genes	95.6	65.94	73.25	56.12	67.37	68.69	65.88
PARSE-new	-	84.38	86.88	71.25	76.50	88.88	80.00

As suggested by Xia et al. (2024), clustering accuracy alone may not suffice for evaluating cell subpopulation identification, as the ground truth can be subject to measurement errors. Therefore, we also evaluate different methods using the Ratio of Global Unshifted Entropy (ROGUE, Liu et al. (2020)), a cell cluster purity metric. ROGUE ranges from 0 to 1, with higher values indicating better clustering results in identifying cell subpopulations. The ROGUE values for the four cell types identified by PARSE, CHIME, SEURAT, and SINCERA are shown in Figure 5(f). PARSE consistently outperforms the other methods based on ROGUE statistics across all cell subpopulations, though its performance on identifying PP cells is slightly lower due to their small representation in the data.

To further visualize the clustering results, we use t-SNE (Van der Maaten and Hinton, 2008; Linderman et al., 2019) to construct a two dimensional embedding of the data, calculated using the PARSE-genes. In the t-SNE plots, data are labeled with the true labels and the labels identified by PARSE, SEURAT, SINCERA, and CHIME, as shown in Figure 5(a). The results clearly demonstrate that PARSE achieves the best cell subpopulation identification. Most of PARSE’s mis-clustering errors are due to the small PP cell population. SEURAT, as shown in Figure 5(c), fails to separate α - from PP cells, leading to many α -cells mistakenly identified as PP cells. Additionally, SINCERA and CHIME fail to separate α - and β - cells.

¹<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE81608>



α vs β	α vs δ	α vs PP	β vs δ	β vs PP	δ vs PP
220	159	142	148	163	126

(c) Numbers of PARSE-genes distinguishing each pair of the four cell subpopulations.

Fig 5: (a) t-SNE plots and ROGUE statistics of HIC data. Cell types are labeled as orange for α -cells, green for β -cells, purple for δ -cells, and cyan for PP cells. Plot (i) uses true labels, Plots (ii)-(v) use results from PARSE, SEURAT, SINCERA, and CHIME. Plot (vi) displays ROGUE statistics for four methods. (b) The heat map of estimated cluster means for PARSE-genes for distinguishing α - and β -cells: genes identifying α -cells are labeled in red with $\times$, β -cells in blue with $\circ$, and δ -cells in magenta with $*$. (c) The numbers of informative pairs among four cluster means of the HIC data.

8.2. Identification of cell-type-specific genes using PARSE. To illustrate how PARSE distinguishes α -cells from β - and δ -cells, the right panel of Figure 5 displays the estimated expression levels of the top 30 genes showing the largest pairwise differences between estimated cluster means. Due to space limitations, gene expression comparisons for all pairs of the four cell subpopulations are provided in Figure S.13 of the supplementary file.

Genes that distinguish α -cells from the other subpopulations, i.e., genes whose expression in α -cells differs from the other three cell types, are highlighted in red, such as CAML (calcium modulating ligand) and LAMB1 (laminin-subunit beta 1). Notably, LAMB1 is a top PARSE-identified gene and is consistently upregulated in α -cells relative to other cell types. It is an α -specific transcription factor, and plays a critical role in pancreatic islet cell development (Miettinen et al., 2000). Genes whose expression differs only between α - and β -cells, such as DSG2 (desmoglein 2) and ILK (integrin-linked kinase), are shown in black. Likewise, genes distinguishing β -, δ -, and PP-cells from the remaining cell types are highlighted in blue, magenta, and brown, respectively. Some identified genes discriminate multiple cell types, whereas others are specific to a single pair. *This reflects our motivation to identify features for clustering at different levels of resolution.* Due to space constraints, additional biological discussion on PARSE-identified genes is deferred to Supplementary Section E. Here, the top 30 genes are presented as “in-or-out” feature selection summary rather than as formal significance assessments. As with other regularization-based methods such as the LASSO,

PARSE provides feature selection results but does not directly yield statistical significance or feature-importance measures. Possible extensions for inference based on PARSE-selected features are discussed in Section 9.

9. Discussion. Motivated by the real-world challenge of identifying cell-type marker genes that characterize unknown cell subpopulations using scRNA-seq data, the central problem is to simultaneously identify informative features from a growing set of candidates that define specific, yet unknown, pairs of clusters. In this paper, we developed the *pairwise reciprocal fusion penalty* (PARSE), which imposes large penalties on small differences between clusters to enhance model-based clustering. Theoretically, we demonstrate that PARSE consistently identifies the true informative features for each pair of clusters.

Several directions remain open for extending the proposed PARSE framework. From a modeling perspective, an important avenue for future research is to generalize the current formulation by incorporating semiparametric or nonparametric components into the mixture model. Such extensions would further relax parametric assumptions on component distributions and enhance the flexibility of the framework in complex data settings.

At the same time, the current version of PARSE has several limitations. Although PARSE is not strictly restricted to this class in practice, its performance may deteriorate when the underlying distribution deviates substantially from Gaussian-like behavior. Extending PARSE to more general frameworks, such as mixtures of zero-inflated Poisson or negative binomial distributions for counting data, is therefore an important direction for future research. A second limitation concerns computational scalability. Although the ℓ_0 -type penalty employed in PARSE enjoys attractive theoretical properties, it can be computationally demanding. One possible extension is to replace the ℓ_0 -penalty with alternative nonconvex penalties, such as the minimax concave penalty (MCP; Zhang, 2010).

In large-sample regimes, PARSE achieves feature-selection consistency at the optimal convergence rate. However, as a penalization-based approach rather than a hypothesis-testing procedure, PARSE does not directly control statistical power or false discovery. In extreme cases, for example, when no meaningful cluster structure is present in the data ($K = 1$), PARSE may face challenges. One possible remedy is to combine PARSE with recently developed post-clustering inference methods, such as the synthetic data-based approach ClusterDE (Song et al., 2023), to further assess and guard against potential inflation of false discoveries. In addition, PARSE does not directly provide measures of statistical significance or feature importance. Nevertheless, one may construct an ad hoc ranking criterion based on the estimated model parameters.

Beyond the specific clustering setting studied here, we believe that the adaptive EM framework underlying PARSE has broader applicability. In principle, any latent-variable model estimated via the EM algorithm may benefit from adaptively restricting attention to a tentative set of “currently active” features during the E- and M-steps. For instance, in certain missing-data problems, the imputation step may depend on a very high-dimensional set of candidate predictors. Allowing the algorithm to dynamically focus on a data-driven subset of relevant variables can substantially reduce the effective dimension and stabilize intermediate updates.

Code. Additional information for the HIC dataset and the data, along with reproducible R codes for the simulation studies, are available at <https://github.com/lyououzhang/parse>.

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